

Survey of the Maximum Edge-Colorable Subgraph Problem

Rahul Drabit Chowdhury*

September 25, 2025

Abstract

The **Maximum Edge-Colorable Subgraph (MECS)** problem is a fundamental challenge in graph theory, aiming to find the largest subgraph that can be properly colored with k colors, with applications spanning network design, computational biology, and optimization. This survey provides a comprehensive overview of the theoretical underpinnings, algorithmic approaches, and fixed-parameter tractability (FPT) results for the MECS problem. We discuss key complexity results, including the FPT parameterizations with respect to vertex-cover number, solution size, matching size, and other structural graph parameters. Special attention is given to the case of **Maximum 2-Edge-Colorable Subgraph (M2ECS)**, which is particularly relevant in practical applications. The survey also highlights recent advancements in kernelization, approximation algorithms, and heuristic approaches. Finally, we explore the real-world applications of MECS in areas such as telecommunication network optimization and graph-based machine learning. By synthesizing these results, this paper aims to provide a broad understanding of the state-of-the-art in edge-coloring problems, offering insights for researchers and practitioners working with large, complex graphs.

1 Introduction

In graph theory, an edge coloring of a graph assigns colors to edges so that adjacent edges have distinct colors. The minimum number of colors needed to edge-color a graph G is called the chromatic index. Vizing's theorem says that any simple graph with maximum degree $\Delta(G)$ can be edge-colored using at most $\Delta(G) + 1$ colors. Deciding whether the chromatic index of a cubic (3-regular) graph is Δ or $\Delta + 1$ is NP-hard.

The maximum edge-colorable subgraph (MECS) problem asks: given a graph G and integers ℓ (the desired number of edges) and p (the number of colors), does G contain a subgraph with at least ℓ edges that is p -edge-colorable? When $p = 1$, the problem becomes finding a maximum matching. For $p \geq 2$, the problem is NP-hard and APX-hard. MECS and its variations are motivated by applications such as call admittance in telecommunication networks, where edges represent calls and colors represent channels, and social network analysis via the strong triadic closure problem. Approximation algorithms also have implications for packing and fault-tolerant guarding problems.

2 Problem Definition and Complexity

Given an undirected graph $G = (V, E)$, a p -edge coloring is a function $\psi : E \rightarrow \{1, \dots, p\}$ such that adjacent edges receive different colors. The maximum edge-colorable subgraph problem is formally defined as follows:

- **Input:** a graph G and integers ℓ and p .
- **Output:** a subgraph $H \subseteq G$ with $|E(H)| \geq \ell$ together with a p -edge coloring of H , or a report that no such subgraph exists.

MECS generalizes maximum matching. When $p = 1$, a 1-edge-colorable subgraph is a matching; thus MECS reduces to finding a maximum matching. Feige, Ofek, and Wieder proved that MECS is NP-hard even for $p = 2$. Because of this hardness, polynomial-time approximation algorithms are important. For simple graphs, the best-known approximation ratios for MECS depend on p : a 6-approximation for

*Department of Computer Science and Engineering, Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh. Email: rahuldrabit@gmail.com

$p = 2$ (Kosowski, 2009), a 7-approximation for $p = 3$ (Rizzi, 2009), and a general $k + 1$ -approximation for $p = k$ using Vizing’s theorem. For multigraphs, Feige et al. (2002) obtained a 13-approximation for $p = 2$ and a 3-approximation for general p based on Shannon’s theorem.

The NP-hardness of MECS implies that achieving a $(1 + \epsilon)$ -approximation is impossible unless $P = NP$. In addition, MECS is APX-hard even when the input is restricted to simple graphs and $p = 2$.

3 Relationship to Related Problems

3.1 Edge Coloring

The MECS problem arises naturally from edge coloring. Vizing’s theorem states that $\chi'(G) \leq \Delta(G) + 1$, where $\chi'(G)$ is the chromatic index. MECS seeks to maximize the number of edges that can be colored using p colors.

3.2 Maximum Matching

When $p = 1$, MECS reduces to finding a maximum matching (each matching is a 1-edge-colorable subgraph).

3.3 k-Matching

For general p , a simple approximation strategy is to compute a k -matching (a subgraph whose maximum degree is k). Then color it and take the union of the k largest color classes. This yields an approximation ratio of $\frac{k}{k+1}$.

3.4 Strong Triadic Closure (STC)

MECS relates to STC, a social network problem where edges have strong or weak labels. The goal is to color edges so that any induced path of length two (a triad) has at least one weak edge. In MECS, edges are deleted to achieve a proper p -edge coloring; in STC, edges are labeled weak or strong rather than deleted. The STC problem and its multi-color variant have applications in detecting strong relationship types in social networks.

4 Definitions and Annotations

4.1 Fixed-Parameter Tractability (FPT)

A computational problem is said to be **fixed-parameter tractable (FPT)** with respect to a parameter k if it can be solved in time $O(f(k) \cdot n^{O(1)})$, where n is the size of the input, and $f(k)$ is some function of the parameter k alone. This means that for small values of k , the problem can be solved efficiently even for large n , while for larger k , the problem may become computationally hard.

4.2 Kernelization

A **kernelization** is a preprocessing technique that reduces the input graph to a smaller equivalent graph (called the kernel) in polynomial time, such that the size of the kernel depends only on the parameter k , not on the overall size of the input. If a problem admits a kernelization, it means the problem can be solved by first reducing the problem to a small kernel and then solving the reduced instance.

4.3 Approximation Algorithms

An **approximation algorithm** for an optimization problem is an algorithm that finds a solution close to the optimal solution in polynomial time. For the MECS problem, approximation algorithms are essential due to its NP-hardness. The goal is to find a solution that is within a specified factor of the optimal solution (e.g., a 6-approximation means the solution will have at most 6 times the number of edges of the optimal solution).

5 Approximation Algorithms and Parameterized Complexity of MECS

5.1 Simple Approach and Vizing-Based Approximations

A classical approximation algorithm due to Feige et al. computes a k -matching F and then uses Vizing's theorem to color F with $k + 1$ colors. The algorithm returns the union of the k largest color classes; the resulting subgraph has at least $\frac{k}{k+1} \cdot |F|$ edges. Since $|F| \geq$ the optimum MECS solution, this yields a $\frac{k}{k+1}$ -approximation for simple graphs. This bound cannot be improved in general because in the complete graph K_{k+1} , any k -edge-colorable subgraph has at most $\frac{k}{k+1} \cdot |E(K_{k+1})|$ edges.

5.2 Improved Ratios for Small Numbers of Colors

For small values of p , better approximation algorithms are known:

- **2-Edge-Colorable Subgraphs:** In simple graphs, Kosowski (2009) gave a 6-approximation. For multigraphs, Feige et al. provided a 13-approximation. Chen and Tanahashi improved the ratio for simple graphs through an LP-based algorithm. They showed that Feige's APX-hardness result was motivated by call admittance in satellite-based telecommunication networks and presented a randomized algorithm that can be derandomized, improving the trivial $4/5$ approximation ratio for MAX EDGE 2-COLORING.
- **3-Edge-Colorable Subgraphs:** Rizzi (2009) designed a 7-approximation for simple graphs. Kowalik and Kamiński later improved the ratio to 15 and provided a 9-approximation for multigraphs.
- **4-Edge-Colorable Subgraphs:** Kowalik and Kamiński obtained an 11-approximation for simple graphs.

These improved algorithms often refine the k -matching approach by improving lower bounds on the optimal solution or by more sophisticated coloring steps. Many algorithms use heuristics such as removing edges from odd cycles and applying randomization or LP relaxation.

5.3 Weighted and Other Variations

In weighted versions, edges have weights, and the goal is to maximize total weight rather than the number of edges. Feige et al. showed that the greedy maximum-coverage algorithm gives an approximation ratio of $1 - (1 - 1/t)^t$ for MAX EDGE t -COLORING. Further work by Kosowski et al. used approximation algorithms for MECS to solve packing and fault-tolerant guarding problems.

5.4 Parameterized Complexity and Kernelization

While MECS is NP-hard in general, it admits fixed-parameter tractable (FPT) algorithms under various parameterizations:

- **Parameterized by Vertex Cover:** Agrawal et al. showed that MECS is FPT when parameterized by the size of a minimum vertex cover of G . Their algorithm guesses the subgraph induced by the vertex cover, sets up an integer linear program (ILP) to choose the remaining edges, and solves the ILP using known FPT techniques.
- **Parameterized by Solution Size ℓ :** They provided two algorithms for MECS when parameterized by the number of edges in the desired subgraph. A deterministic algorithm runs in $O^*(4\ell + o(\ell))$ time and uses color-coding and divide-and-color techniques, while a randomized algorithm runs in $O^*(2\ell)$ time via reduction to the rainbow matching problem.

- **Distance-From-Triviality Parameterization:** Grüttemeier et al. studied the decision version of MECS where one asks if at most k edges can be deleted to obtain a graph with maximum degree $c - 1$. They introduced the parameter ξ_{c-1} , the minimum number of edges that need to be deleted to reduce G to maximum degree $c - 1$. For fixed c , they obtained a linear kernel with at most $4\xi_{c-1} \cdot c$ vertices.
- **Other Structural Parameters:** When p is fixed (e.g., $p = 2$), Aloisio and Mkrtchyan showed that MECS is FPT with respect to parameters such as path-width, carving-width, and cycle space dimension. Galby et al. showed that the edge coloring problem is FPT when parameterized by the number of colors and the number of vertices of maximum degree.

5.5 Kernelization Lower Bounds

Agrawal et al. also proved that the decision version of MECS admits a polynomial kernel of size $O(k \cdot p)$ vertices when parameterized by $p + k$, where k can be ℓ , the vertex cover number, or $\ell - mm(G)$ (with $mm(G)$ denoting the maximum matching size). They further showed that, under standard complexity assumptions, the problem does not admit a kernel of size $O(k^{1-\epsilon} \cdot f(p))$ for any $\epsilon > 0$.

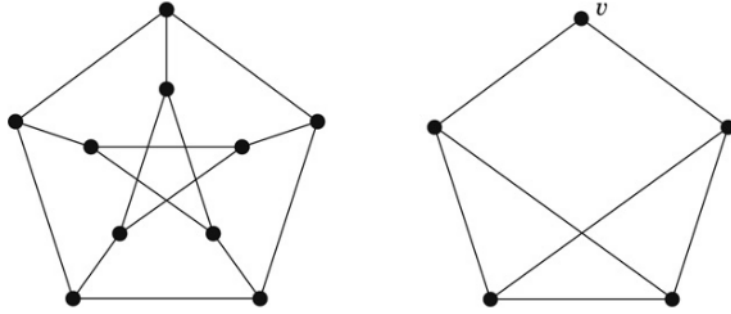


Figure 1: The Petersen graph (on the left) shows that $\frac{13}{15}$ in Corollary 2 is tight. Graph G_5 (on the right) shows that $\frac{6}{7}$ in Theorem 3 is tight.

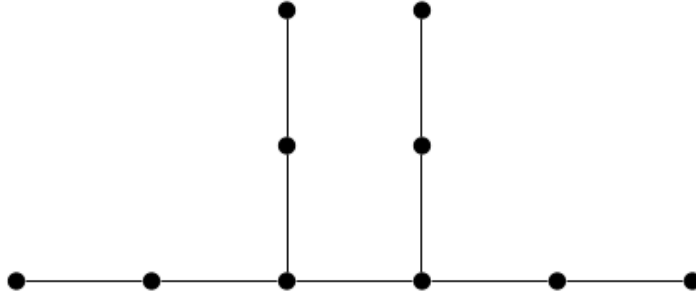


Figure 2: A graph illustrating a perfect matching and 2-edge-colorability. The edges can be colored using two colors, ensuring that no two adjacent edges share the same color, while each vertex is matched with exactly one other vertex in the perfect matching.

6 Summary of Algorithms and Parameterized Results

The tables below consolidate the approximation ratios and parameterized algorithms described above.

6.1 Approximation Ratios for p -Edge-Colorable Subgraphs

Graph Class	Colors p	Approximation Ratio
4 Simple graphs	2	6-approximation
	3	7-approximation
	3	15-approximation (improved)
	4	11-approximation
	k	$\frac{k}{k+1}$ -approximation via Vizing's theorem
3 Multigraphs	2	13-approximation
	k	3-approximation (Shannon & Feige)
	k	$\xi(k)$ -approximation, where $\xi(k) \rightarrow 1$ as $k \rightarrow \infty$

Table 1: Approximation Ratios for p -Edge-Colorable Subgraphs

6.2 Summary of Parameterized Results and Kernels

Parameterization	Result	Key Ideas
Vertex cover number $\nu_c(G)$	FPT algorithm using ILP	Guess the subgraph induced by the vertex cover and solve an ILP for the remaining edges
Solution size ℓ	Two FPT algorithms: deterministic $O^*(4\ell + o(\ell))$ and randomized $O^*(2\ell)$	Color-coding & divide-and-color; reduction to rainbow matching
$p + k$	Kernel with $O(k \cdot p)$ vertices; no kernel of size $O(k^{1-\epsilon} \cdot f(p))$ unless NP coNP/poly	Kernelization and lower-bound techniques
Distance to max-degree $\leq c - 1$	Linear kernel with $\leq 4\xi_{c-1} \cdot c$ vertices	Remove low-degree vertices; rely on Vizing's theorem
Vertex-deletion distance to components of order $\leq c$	Kernel with $O(c^3 \cdot \lambda_c)$ vertices	Structural reduction to small components
Path-width, carving-width, cycle space dimension (for $p = 2$)	MECS is FPT	Structural decomposition; dynamic programming
Number of colors & number of maximum-degree vertices	Edge coloring is FPT	Parameterization by color count and high-degree vertices

Table 2: Summary of Parameterized Results and Kernels

7 Applications

MECS and related problems arise in several applied settings:

7.1 Telecommunication Networks

Feige, Ofek, and Wieder introduced MECS while studying call admittance in satellite-based telecommunication networks, where each call occupies a channel (color). They observed that the problem is APX-hard even for simple graphs and $p = 2$.

7.2 Social Network Analysis

The strong triadic closure problem is used to discover strong ties in social networks. In the multi-STC problem, edges are colored as weak or with one of c strong colors. The parameterized kernelization results for MECS extend to multi-STC, leading to algorithms that find these structures in large networks.

7.3 Packing and Guarding Problems

Approximation algorithms for MECS have been applied to packing problems and fault-tolerant guarding problems. Kosowski et al. showed how approximations for MAX EDGE 2-COLORING can be used to derive approximations for such combinatorial optimization problems.

8 Open Problems and Future Directions

The MECS problem remains challenging. Open directions include:

- **Improved Approximation Ratios:** Despite decades of work, tight bounds on the approximation ratio for general p are not known. The ratio $\frac{k}{k+1}$ for simple graphs is essentially optimal for complete graphs, but there may be room for improvement on restricted graph classes or average-case instances.
- **Weighted and Capacity-Constrained Variants:** Many practical applications involve capacities or weights on edges. Extending parameterized and approximation results to these variants is an open research direction.
- **Dynamic and Online MECS:** In real-time networks, calls or social ties may change over time. Designing online algorithms that maintain large edge-colorable subgraphs under updates is largely unexplored.
- **Parallel and Distributed Algorithms:** For large-scale graphs such as social networks, parallel or distributed algorithms for MECS or its kernelization would be valuable.

9 Conclusion

The maximum edge-colorable subgraph problem provides a rich interplay between graph theory and algorithm design. It generalizes maximum matching and is closely connected to edge coloring. Research has produced a variety of approximation algorithms ranging from simple k -matching heuristics to sophisticated LP and randomized techniques. On the parameterized side, the problem admits FPT algorithms for parameters that measure the solution size, vertex cover number, or distance from trivial instances, but it remains intractable without parameterization. Applications in telecommunication networks and social network analysis underscore the practical significance of MECS. Continued research on approximations, parameterized complexity, and kernelization promises further progress on this fundamental problem.

Acknowledgments

I would like to thank my professor, [Dr. Md. Saidur Rahman], for providing guidance and feedback throughout the course of this project. Their insights were invaluable in shaping the content of this survey.

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